# altairsim — Quick Reference

2026-09-04 · aeab8b6

## Getting out, and back in

| Key | Does |
|---|---|
| `^E` | **ATTN —** stop the machine and take the keyboard back. Nothing is lost. |
| `RUN` | Resume, at the exact instruction it stopped on. |
| `QUIT` | Leave. (There is no `EXIT`.) |

## Editing the command line

`Tab` completes what you are typing — a command, then a board id, then its property names, then a property's values (`SET mem0 fill=` then `Tab`); a second `Tab` lists the choices. `Up` / `Down` walk the command history, saved per directory in `.altairsim_history`.

| Key | Does |
|---|---|
| `Ctrl-A` / `Ctrl-E` (or `Home` / `End`) | start / end of line |
| `Alt-B` / `Alt-F` (or `Ctrl-Left` / `Ctrl-Right`) | back / forward one word |
| `Ctrl-W` / `Ctrl-K` / `Ctrl-U` | erase word behind / to end of line / whole line |
| `Backspace` / `Delete` | erase before / under the cursor |

## Command line

```
altairsim [options] [machine]

  machine            a built-in name, or a config file (has a '/' or ends .toml).
                     Omitted: ./altairsim.toml if there is one, else `default`.
  -m, --machine <n>  ALWAYS a built-in name -- never a file.
  -f, --file <path>  ALWAYS a file -- never a built-in name.
  -n, --none         empty backplane: no boards, no memory, nothing.
  -l, --list         list the built-in machines and exit.
  -s, --script <f>   run a command script, then exit with its status.
  -x, --exec <cmd>   run one monitor command (repeatable), then exit.
  -i, --interactive  after --script/--exec, stay in the monitor.
      --mcp          MCP server on stdio.
  -v, --version      print the version and exit.
  -h, --help         print this help and exit.
```

## Monitor commands

Type the part before the bracket.

<table>
<tr><th>Command</th><th>Does</th><th>Usage</th></tr>
<tr><td>BO[ARDS]</td><td>List, add, or remove boards on the backplane.</td><td>BOARDS [LIST]|ADD <type> <id> [k=v...]|REMOVE <id></td></tr>
<tr><td>B[REAK]</td><td>Set a breakpoint on an address, memory/I/O access, or tape stop.</td><td>BREAK [<addr> | MEM R|W <addr> | IO R|W <port> | TAPE STOP] [IF <expr> | LOADS <expr>] [TRACE ON|OFF]</td></tr>
<tr><td>COM[PARE]</td><td>Compare a range of memory against another address.</td><td>COMPARE <range> <addr></td></tr>
<tr><td>C[ONFIG]</td><td>Load or save the whole machine as a TOML file.</td><td>CONFIG LOAD <f.toml> | CONFIG SAVE <f.toml></td></tr>
<tr><td>CONN[ECT]</td><td>Attach a serial unit to an endpoint (console, socket, file, ...).</td><td>CONNECT <id>:<u> <endpoint></td></tr>
<tr><td>CONS[OLE]</td><td>Show or set the host console's properties.</td><td>CONSOLE [<k>=<v>...]</td></tr>
<tr><td>DE[POSIT]</td><td>Write bytes into memory at an address.</td><td>DEPOSIT <addr> <bytes...></td></tr>
<tr><td>DI[SASM]</td><td>Disassemble memory into instructions.</td><td>DISASM [<addr>|<range>] [n] [CPU=8080]</td></tr>
<tr><td>DISC[ONNECT]</td><td>Unplug the endpoint from a serial unit.</td><td>DISCONNECT <id>:<u></td></tr>
<tr><td>D[UMP]</td><td>Show memory as hex and ASCII.</td><td>DUMP [<addr>|<range>] [WIDTH=16]</td></tr>
<tr><td>E[DIT]</td><td>Enter bytes into memory interactively from an address.</td><td>EDIT <addr> [ROM]</td></tr>
<tr><td>EX[AMINE]</td><td>Point the front panel at an address (and show that byte).</td><td>EXAMINE [<addr>]</td></tr>
<tr><td>F[ILL]</td><td>Fill a range of memory with a byte.</td><td>FILL <range> <byte></td></tr>
<tr><td>HE[LP]</td><td>Show help for a command.</td><td>HELP [<command>]</td></tr>
<tr><td>H[ISTORY]</td><td>Replay the recent instruction (or bus-cycle) history.</td><td>HISTORY [BUS|CPU] [n]</td></tr>
<tr><td>I[N]</td><td>Read a byte from an I/O port.</td><td>IN <port></td></tr>
<tr><td>L[OAD]</td><td>Load a file into memory (binary, Intel hex or S-record).</td><td>LOAD <file> [AT <addr>] [FORMAT=BIN|HEX|SREC] [ROM]</td></tr>
<tr><td>M[OUNT]</td><td>Put a disk or tape image into a drive; a .imd is converted to a raw .dsk beside it.</td><td>MOUNT <id>[:<u>] <file> [WP] [CREATE] [extract[=<base>]] [k=v...]</td></tr>
</table>

<table>
<tr><td>MOV[E]</td><td>Copy a range of memory to another address.</td><td>MOVE <range> <dest> [ROM]</td></tr>
<tr><td>N[EXT]</td><td>Step one instruction, running any CALL/RST to completion.</td><td>NEXT</td></tr>
<tr><td>NO[BREAK]</td><td>Remove a breakpoint, or all of them.</td><td>NOBREAK [id]</td></tr>
<tr><td>O[UT]</td><td>Write a byte to an I/O port.</td><td>OUT <port> <byte></td></tr>
<tr><td>P[OWER]</td><td>Power-cycle the machine -- the only thing that clears RAM.</td><td>POWER</td></tr>
<tr><td>Q[UIT]</td><td>Leave the simulator.</td><td>QUIT</td></tr>
<tr><td>REGI[ON]</td><td>Add a RAM or ROM region to a memory board.</td><td>REGION ADD <id> type=ram|rom at=<addr> [size=|mount=]</td></tr>
<tr><td>RE[GS]</td><td>Show the CPU registers (SET REG changes one).</td><td>REGS | SET REG <r>=<v></td></tr>
<tr><td>RES[ET]</td><td>Reset the machine, keeping RAM (RESET CPU resets just the processor).</td><td>RESET [CPU]</td></tr>
<tr><td>REST[ORE]</td><td>Load machine state back from a snapshot.</td><td>RESTORE <file></td></tr>
<tr><td>R[UN]</td><td>Start or resume the machine, optionally at an address.</td><td>RUN [addr]</td></tr>
</table>

<table>
<tr><td>SA[VE]</td><td>Write a range of memory out to a file.</td><td>SAVE <file> <range> [FORMAT=BIN|HEX|OCTAL|PRN]</td></tr>
<tr><td>SEA[RCH]</td><td>Find bytes or a string in a range of memory.</td><td>SEARCH <range> <bytes...>|"str"</td></tr>
<tr><td>SE[T]</td><td>Change a property of a board, the console, display, a register, or the bus.</td><td>SET <id>[:<u>]|CONSOLE|DISPLAY|REG|BUS <k>=<v></td></tr>
<tr><td>SH[OW]</td><td>Display the state of a board, the bus, or the machine.</td><td>SHOW <id>|BOARDS|BOARD <type> [UNITS]|MACHINES|MACHINE [<name>]|BUS [MAP|IO|IRQ|CONTENTION]|ROMS|MOUNTS|PATHS|CONSOLE|DISPLAY|SYMBOLS|VERSION</td></tr>
<tr><td>SN[APSHOT]</td><td>Save the whole machine state to a file.</td><td>SNAPSHOT <file></td></tr>
<tr><td>S[TEP]</td><td>Run one instruction (or n), showing the registers after each.</td><td>STEP [n]</td></tr>
<tr><td>SY[MBOLS]</td><td>Load or clear a symbol table for disassembly.</td><td>SYMBOLS LOAD <file> [REPLACE] | SYMBOLS CLEAR</td></tr>
<tr><td>T[RACE]</td><td>Log every bus cycle while the machine runs.</td><td>TRACE ON|OFF [file] [MASK=IN,OUT,IRQ,DMA,CONTENTION]</td></tr>
<tr><td>TY[PE]</td><td>Feed text to the guest as if typed at its keyboard.</td><td>TYPE "text"</td></tr>
<tr><td>U[NMOUNT]</td><td>Take a disk or tape out of a drive.</td><td>UNMOUNT <id>:<u></td></tr>
</table>

| | | |
|---|---|---|
| `W[HO]` | Say which board answers an address or I/O port. | `WHO <addr> \| WHO IO <port>` |

# Boards

## CPU

| Type | What it is |
|---|---|
| `6800` | Altair 680b CPU board: a Motorola 6800 at 500 KHz. Decodes nothing -- it drives the bus. The 88-CPU's twin, one core down, with memory-mapped I/O |
| `8080` | MITS 88-CPU: an 8080A at 2 MHz. Decodes nothing -- it drives the bus |
| `8085` | Generic 8085 CPU board. Decodes nothing -- it drives the bus. The 88-CPU's twin, with an 8085 core (RIM/SIM + TRAP/RST 5.5/6.5/7.5) |
| `z80` | Generic Z80 CPU board. Decodes nothing -- it drives the bus. The 88-CPU's twin, with a Z80 core |

## Memory

| Type | What it is |
|---|---|
| `bankmem` | S-100 bank-switched RAM. One card, four decoders (card=vector\|cromemco64kz\|northstar\|expandoram2): a write-only select port swaps which RAM plane(s) drive the bus. Each card owns its own decode -- one-hot select (Vector 40), 8-bit bank mask (Cromemco 40), on/off+one-hot toggle (North Star C0), or PROM page-select (ExpandoRAM II FF, approximated) |
| `memory` | RAM/ROM board: a list of regions and PHANTOM* -- plain, unbanked memory (bank switching is its own board, `bankmem`) |
| `v2z80rom` | S100Computers V2 Z80 CPU board -- its onboard paged monitor EEPROM (the Z80 itself is board 'z80cpu'). An 8K 28C64 at F000-FFFF holding two 4K pages, builtin:master0 (low) / master1 (high), selected by OUT D3H bit1 (bit0=1 inactivates the EEPROM so RAM shows through). Shadows RAM in its window while enabled. Cold-start the MASTER monitor with startup=["RUN F000"]; the 'I' command boots CP/M 3 off a dualsd card |

### Disk

| Type | What it is |
| --- | --- |
| `16fdc` | Cromemco 16FDC: WD FD1793 soft-sector floppy (single + double density), up to 4 drives. Disk registers at 30-34, a TMS 5501 console UART at 00-09 (unit 'tty'), and a 4K RDOS 2.52 boot PROM at C000 (OUT 40H banks it out, RESET restores it). Boots CDOS |
| `64fdc` | Cromemco 64FDC: the 16FDC's 1983 successor -- same FD1793 + TMS 5501, carrying an 8K RDOS 3.12 boot PROM at C000-DFFF (OUT 40H banks it out, RESET restores it). Boots CDOS |
| `dcdd` | MITS 88-DCDD: 8" hard-sector floppy, up to 16 drives. Three ports at BASE+0..2. INVERTED status bits |
| `dualide` | S100Computers IDE-AB (CF): the IDE/CompactFlash half of the IDE+ESP32 combination board -- two CF sockets (drives 0/1 = A:/B:) for CP/M 3. Five 8255 ports at BASE+0..4 (default 30): A/B data, C control lines, mode config, drive select. Programmed-I/O ATA register engine (LBA read/write, 512-byte sectors). No boot PROM -- the CPU board's monitor boots CP/M from the CF. Mounts the SAME card image as dualsd (a .img with a .geo geometry sidecar); pair with dualsd for the full A:/B:+C:/D: system |
| `dualsd` | S100Computers Dual SD: two microSD sockets (drives 0/1) presented as raw 512-byte-sector CF/SD cards, for CP/M 3. Two ports at BASE+0..1 (default 80): status/command + data. Programmed-I/O command/handshake engine (33H-lead + 8 commands). No boot PROM -- the CPU board's monitor loads CP/M from track 0. Mount a card image (a .img with a .geo geometry sidecar) |
| `hdsk` | MITS 88-HDSK Datakeeper: Pertec hard disk, 256-byte sectors from a linear .DSK. Eight ports at BASE+0..7 (default A0). Command/handshake protocol, four page buffers |
| `icom` | iCOM FD3712/FD3812 8" floppy: a programmed-I/O command/handshake controller on the S-100 Interface board. Two ports at BASE+0..1 (default C0) plus a boot PROM and 6810 scratch RAM in high memory (rom=builtin:icom-fd3712-cpm \| icom-fd3712-fdos \| icom-fd3812-cpm). Boots CP/M 2.2 (single and double density) and FDOS. Up to 4 drives |
| `mds` | MITS 88-MDS: 5.25" minidisk, 4 drives. Same three ports as the dcdd -- but 300 RPM, 64 us/byte, and a motor that stops after 6.4 s |
| `tarbell` | Tarbell #1011: single-density WD FD1771 floppy, up to 4 drives. Eight ports at BASE+0..7 (default F8). Carries a 32-byte boot PROM that shadows 0000 over PHANTOM* -- boots CP/M automatically at reset (bootstrap=on) |
| `tarbelldd` | Tarbell #2022: double-density WD FD1791 floppy (mixed-density media, SD track 0), up to 4 drives. The single-density card's twin with a bitmap OUT-FC latch and a port-FD DMA/ext-addr register. Same 32-byte boot PROM |
| `versafloppy` | SD Systems VersaFloppy I/II: WD FD177x soft-sector floppy, up to 4 drives. Eight ports at BASE+0..7 (default 60). variant=vfi (FD1771, single density) \| vfii (FD1791, single+double). Boots SDOS with the SBC-200 + DDBIOS |

## Serial

| Type | What it is |
| --- | --- |
| `2sio` | MITS 88-2SIO: two 6850 ACIAs, units 'a' and 'b'. Four ports at BASE+0..3 |
| `680io` | Altair 680b onboard I/O: a 6850 ACIA console ('tty') at F000/F001 and the config-strap read port at F002. Memory-mapped |
| `gsio` | Generic SIO: a strap-configurable serial board with TWO independent channels, units 'a' and 'b' (configure each under [board.unit.a] / [board.unit.b]). Basic transmit/receive only -- no programmable word length, parity, stop bits or framing/overrun status; a specific card that needs those is a separate emulated board. Per channel you strap: a status/control port (status_port -- read synthesizes DAV/TBMT at bit positions you pick, write is discarded) and a data port (data_port); one inverter_gate knob inverts both status bits together. Built-in profiles preset the straps: profile=sior0 (MITS SIO Rev 0, the default) \| tuart (Cromemco TU-ART) \| imsai-sio2 \| compupro-if2 (CompuPro Interfacer II) \| compupro-ss1 (CompuPro System Support 1). Channel a defaults to ports 0/1, b to 2/3. Polled, no interrupts. CONNECT each channel to a file, socket, serial port, in:/out: |
| `io4` | SSM IO-4 (2P+2S): the real Solid State Music board -- two full-duplex serial channels AND a four-port parallel section. Serial units 'a'/'b' are real 1602-family UARTs with programmable word length (data_bits 5-8), parity and stop bits (unlike the generic gsio), plus the full status-word strap-up (stat_* map, invert_status, port_reversal) and named profiles (default altair-rev1). A 4-port block set by switch S3 (default 0-3): Serial A status/data at BASE+0/+1, Serial B at BASE+2/+3. Parallel units 'pa'/'pb' are 8212 latched ports (input latch + service request, output latch) on their own 2-port block set by switch S4 (par_port, default 4-5): Parallel A at PAR+0, B at PAR+1; a byte the far end sends is strobed in, a write latches out, and the §3.2.2 status/data console flag is strappable (dav_bit/dav_source/dav_active_low). If the two blocks overlap, neither section answers the shared ports. Configure each unit under [board.unit.a] / [board.unit.pa] etc.; CONNECT each to a file, socket, serial port, in:/out:. Interrupts are a separate phase |
| `pmmi` | PMMI MM-103: Bell 103 modem on an S-100 card, unit 'line'. Four ports at BASE+0..3 (default C0), read/write different registers. Transmit/receive over a ByteStream; CONNECT it to in:/out: files. No dialer; modem status is a fixed stub |
| `propio` | S100Computers Console IO Board (Parallax-Propeller console), unit 'serial'. A strap-configurable serial subtype preset to the board's documented convention: status/data at 00/01, RX-ready = status bit 1, TX-ready = status bit 2, both active high. Every strap (status_port/data_port/dav/tbmt/inverter_gate) is still overridable -- the real board is jumpered. Polled, no interrupts. CONNECT it to a file, socket, serial port, in:/out: |
| `sbc` | SD Systems SBC-100/200: Z80 single-board computer. One 8-port block (78-7F): Intel 8251 console (unit 'tty', data 7C / status 7D, RxD->/DSR auto-baud for MSMONR21), Z80-CTC (78-7B) whose ch1 raises a mode-2 keyboard interrupt (vector 0x82) off the 8251 RxRDY, and a parallel port (7E/7F) whose OUT 7F bit 1 switches the onboard PROM out. Optional onboard boot PROM via [[board.socket]] (at+mount). variant=sbc100\|sbc200 |
| `sio` | MITS 88-SIO: one COM2502 UART, unit 'tty'. Two ports at BASE+0..1. INVERTED status bits |
| `turnkey` | MITS 8800b Turnkey Module: phantom boot PROM (FC00-FFFF), integrated 6850 SIO (unit 'tty', default 0x10), sense switches at FF, and the Auto-Start JMP jam. Sockets via [[board.socket]] |

### Tape

| Type | What it is |
| --- | --- |
| `680kcacr` | Altair 680b KCACR audio-cassette interface: a 1602-family UART recording Kansas City Standard FSK, memory-mapped at F010 (status/control) and F011 (data), active-LOW. Adds software motor control (control D7=on, D6=off) and interrupt-driven transfer (D0/D1 enables pull the 6800 IRQ). Reuses the 88-ACR tape machinery -- MOUNT a tape, WIND/REWIND it |
| `acr` | MITS 88-ACR: cassette. An 88-SIO B + an FSK modem, unit 'tape'. Brings the WIND/REWIND/EXTRACT verbs and a tape counter |
| `uio` | MITS 88-UIO: serial + cassette on one board. A 6850 (unit 'serial', default 0x10) and an 88-ACR cassette section (unit 'tape', default 0x06) with motor control and a SW-1 MITS/Kansas-City modulation switch. Defaults reproduce the standard 0x10 + 0x06 layout |

### Parallel and printer

| Type | What it is |
| --- | --- |
| `4pio` | MITS 88-4PIO: up to four 6820 PIAs, sections ja/jb.. per port. 16 ports from BASE (default 20). Software-set direction; CONNECT each section |
| `680uio` | Altair 680b Universal I/O: a second 6850 ACIA serial port ('serial') and a 6820 PIA parallel port (sections 'p1a/p1b', 'p2a/p2b' with pias=2) in an S9-relocatable window (default base F000: serial F006/F007, PIA F008-F00F), plus fixed switch inputs at F003 and a non-latched output at F010-F013. Memory-mapped, active-high |
| `c700` | MITS 88-C700: Centronics line-printer controller, unit 'prn'. Two ports at BASE+0..1 (default 02). Output-only; CONNECT it to a file, a socket, or a real printer queue |
| `d7a` | Cromemco D+7A: analog + parallel I/O. Eight ports from BASE (default 18): one parallel port + seven two's-complement A/D-in/D/A-out channels. Reads 1-2 JS-1 joysticks from the host |
| `lpc` | MITS 88-LPC: 88-LP line-printer controller, unit 'prn'. Two ports at BASE+0..1 (default 02). Line-buffered: 6-bit codes + PRINT/LINE FEED/CLEAR. CONNECT it to a file, a socket, or a real printer queue |
| `pio` | MITS 88-PIO: 8-bit parallel port, units 'out'/'in'. Two ports at BASE+0..1 (default 04). CONNECT a printer, a keyboard, a socket |

### Video

| Type | What it is |
| --- | --- |
| `dazzler` | Cromemco Dazzler: color graphics from a framebuffer in main RAM. Two ports at BASE+0..1 (default 0E): control/status and format. 32x32 to 128x128, 16 colors/greys. Needs a Display |
| `vdb8024` | SD Systems VDB-8024: an 80x24 video terminal on one board -- the video console for an SBC-100/200 (the alternative to the 8251). Two I/O ports at BASE+0..1 (default 00): status/keyboard/display. Unit 'keyboard' (CONNECT). Optional keyboard-strobe interrupt strap (interrupt=vi0..vi7) for the SBC-200's CTC to vector -- what the SD video CBIOS needs; polled by default. Boots sdmonv21. Needs a Display |
| `vdm1` | Processor Technology VDM-1: memory-mapped 16x64 video, screen RAM at BASE (default CC00), scroll/status port (default CC). Needs a Display |

### Systems

| Type | What it is |
| --- | --- |
| `sol` | Processor Technology Sol-PC I/O: serial, keyboard, parallel, CUTS tape as one board. Seven ports F8..FE. Units serial/printer/keyboard (CONNECT) and tape1/tape2 (MOUNT). Brings the WIND/REWIND/EXTRACT verbs and a tape counter |

### PROM programmer

| Type | What it is |
| --- | --- |
| `pb1` | SSM PB1: 2708/2716 EPROM programmer + on-board EPROM board. A 4K programming-socket window (default D000, sockets U22=2708/U23=2716) and one control port (default 10): OUT arms the board and picks the chip (D0=2708, D1=2716), then a window write burns a byte and a window read disarms it. Save the burn to a host hex file with `SAVE file window`. Optional read-only on-board EPROM area via [[board.prom]] (at + mount). The board ships no firmware -- run any 2708/2716 burner; SSM's own from the PB1 manual is in examples/pb1 |

### Other

| Type | What it is |
|---|---|
| `fp` | Altair front panel: the SENSE switches a guest reads at IN 0FFH -- a configured byte (SET fp0 sense= or TOML), not toggled here. No OUT |
| `hostbridge` | Host Bridge: guest <-> host file transfer, sandboxed. OUR OWN BOARD, not a period one. Two ports at BASE+0..1. R.COM/W.COM/HDIR.COM |
| `ss1` | CompuPro System Support 1: multifunction S-100 board. Dual 8259A interrupt controllers in a master/slave cascade (master/slave at base+0..+3; master watches VI0-6 and drives pin 73, slave takes the timer OUTs and the UART's Rx/TxRDY), an 8253 interval timer (three counters + control at base+4..+7, 2 MHz clock), the OKI MSM5832 battery-backed real-time clock/calendar (command/data at base+10/+11) and a 2651 UART serial channel (base+12..+15); base default 50H. The 9511/9512 math socket is unpopulated |
| `virtc` | MITS 88-VI/RTC: vectored interrupts (VI0-VI7 -> RST n) and a real-time clock. One port at FE |

## Machines

| Machine | What it is |
|---|---|
| `8085` | A minimal 8085 machine: an `8085` CPU, 64K of RAM, and a 2SIO console. |
| `acuter` | ACUTER at F000 -- CUTER on a plain Altair, with a terminal instead of a VDM. |
| `altair680` | The Altair 680b -- MITS's second machine, and a different animal from the 8800. |
| `altmon` | An Altair with a monitor in ROM and a terminal on it. |
| `amon` | AMON 3.1 in a 4K EPROM at F000 -- Martin Eberhard's full-featured Altair monitor. |
| `bankmem` | A bank-switched RAM machine: a Z80, a console, and a Vector Graphic 64K bankmem. |
| `basic4k` | The machine Altair 4K BASIC was sold to run on: an 88-SIO Teletype, a cassette in the ACR. |
| `basic8k` | The machine Altair 8K BASIC was sold to run on: an 88-2SIO terminal, a cassette in the ACR. |
| `cdbl` | The `default` machine with the Combo Disk Boot Loader in the PROM socket. |
| `compupro` | A stock Altair with a CompuPro System Support 1 board for its clock/calendar. |
| `cuter` | CUTER 1.3 driving a Processor Technology VDM-1 -- the real Sol/CUTS monitor. |
| `dazzler` | A Cromemco Dazzler in an Altair -- the bench for the S-100's first color graphics card. |
| `default` | The machine you get when you name none: 56K, and the DBL boot PROM at FF00. |
| `dualide` | S100Computers "IDE-AB CF" machine -- a V2 Z80 CPU board booting CP/M 3 off a CompactFlash card. |
| `dualidesd` | S100Computers "IDE-AB CF+ESP32" machine -- both boards, CP/M 3 on CF drives A:/B: and SD drives C:/D:. |
| `dualsd` | S100Computers "Dual SD" machine -- a V2 Z80 CPU board booting CP/M 3 off a microSD card. |
| `icom` | iCOM FD3712 8" floppy machine -- boots CP/M 2.2 off a single-density iCOM disk. |
| `lineprinter-lpc` | The `default` machine with an 88-LPC line printer at port 02; CONNECT `lpt0:prn` to a file or the console. |
| `lineprinter` | The `default` machine with an 88-C700 line printer at port 02; CONNECT `lpt0:prn` to a file or the console. |
| `minidisk` | The Altair Minidisk: an 88-MDS at 08 and the MDBL boot PROM. You supply the 5.25" disk. |
| `original` | The Altair as it actually left Albuquerque. |
| `parallel` | The `default` machine with two MITS parallel boards: an 88-PIO and an 88-4PIO. |
| `ps2` | The machine MITS Programming System II ran on: basic8k's cards, but not basic8k's bootstrap. |
| `ps2int` | MITS Programming System II, WITH INTERRUPTS. `ps2` with A9 down and an 88-VI/RTC in it. |
| `rombasic` | MITS Extended ROM BASIC 16K -- Extended BASIC that runs directly out of ROM. |

| | |
|---|---|
| `sbc200` | SD Systems SBC-200 -- a 4 MHz Z80 single-board computer running the MSMONR21 monitor. |
| `sbc200v` | The SD Systems SBC-200 with a VDB-8024 VIDEO console: SDMONV21 on an 80x24 screen. |
| `sol20` | The Processor Technology Sol-20 -- an integrated 8080 machine, running SOLOS. |
| `tarbell` | Tarbell #1011 single-density floppy machine -- boots CP/M 2.2 the moment a disk is in it. |
| `tarbelldd` | Tarbell #2022 double-density floppy machine -- boots CP/M 2.2 the moment a disk is in it. |
| `turnkey` | The MITS 8800bt -- an Altair with a Turnkey Module where the front panel used to be. |
| `vdm1` | A Processor Technology VDM-1 in an Altair, and a demo that draws on it. |
| `z80` | A minimal Z80 machine: a `z80` CPU, 64K of RAM, and a 2SIO console. |

## A machine file, in one look

```
[machine]
name    = "mine"
base    = "default"        # start from a machine, and say what is DIFFERENT
startup = ["RUN FF00"]     # the operator's own keystrokes. There is no BOOT verb.

[[board]]                  # type + a NEW id      -> ADD the card
type = "2sio"              # type + an id from the base -> REPLACE it outright
id   = "sio0"              # NO type + an id      -> MODIFY the one already there
port = 10                  # remove = true        -> PULL THE CARD OUT

  [board.unit.a]           # a unit's own settings
  connect = "console"

  [[board.region]]         # a list the card owns (memory)
  type = "ram"
  at   = 0000              # hex: it is an address
  size = "56K"             # decimal: it is a size

  [[board.drive]]          # a list the card owns (disk controllers)
  unit  = 0
  mount = "cpm.dsk"        # relative to THIS FILE

[console]                  # the HOST's terminal -- not a board
strip7out = true
base      = octal          # read/print the wire class in split octal (MITS style)
```

**Paths:** a path *inside* a machine file is relative to **that file**. A path you *type* is relative to **your shell**.

## Endpoints — `CONNECT <id>:<unit> <endpoint>`

| Endpoint | Is |
|---|---|
| `console` | the host terminal. Exactly one unit may hold it. |
| `null` | nowhere. Writes vanish, reads never come. |
| `loopback` | itself — what you write comes back. |
| `scripted` | a caller in place of a human — what MCP and the tests type into. |
| `socket:PORT` | **listens** — this is telnet-in. |
| `socket:HOST:PORT` | **calls out**. |
| `serial:DEVICE` | a real serial port on this host. |
| `in:PATH` | a host file as a reader (paper tape). `?cps=N` paces it. |
| `out:PATH` | a host file as a punch — 8-bit clean, never truncating. |
| `terminal` | a window the simulator draws itself (SDL builds). `?emulation=vt100\|adm3a\|vt52\|h19`, `?size=COLSxROWS`. |
| `printer:QUEUE` | a real print queue on this host. |
| `<endpoint>\|FILE` | a tap: append `\|FILE` to any endpoint to also log the line. |
| `<endpoint>\|socket:PORT` | a live mirror: `telnet` in to watch and take over. `?ro` = watch-only. |